12. Tic Tac Toe Game

```
board = [" "] * 9

def display():
    print(board[0:3])
    print(board[3:6])
    print(board[6:9])

player = "X"

for i in range(9):
    display()
    pos = int(input("Enter position (0-8): "))

    if board[pos] == " ":
        board[pos] = player
        if any(
            all(board[j] == player for j in line)
            for line in [(0,1,2),(3,4,5),(6,7,8),
                         (0,3,6),(1,4,7),(2,5,8),
                         (0,4,8),(2,4,6)]
        ):
            display()
            print(player, "wins!")
            break
        player = "O" if player == "X" else "X"
    else:
        print("Position already occupied")
else:
    display()
    print("Draw!")
```

main.py Visualize Run ▶ Output

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player = "X"

for i in range(9):
    display()
    pos = int(input("Enter position (0-8): "))

    if board[pos] == " ":
        board[pos] = player
        if any(
            all(board[j] == player for j in line)
            for line in [(0,1,2),(3,4,5),(6,7,8),
                         (0,3,6),(1,4,7),(2,5,8),
                         (0,4,8),(2,4,6)]
        ):
            display()
            print(player, "wins!")
            break
```

```
[' ', ' ', ' ']
[' ', ' ', ' ']
[' ', ' ', ' ']
Enter position (0-8):
```